

PAX vibration diagnostics — run report

RUN

203321_pax-vibration_rubber-tip-base

Status: completed | Started: 2026-09-25T19:33:21.126884+00:00

PURPOSE

Compare fast PD variance with the PAX motor running and stopped. Log all fresh PAX telemetry and static sphere-angle variance during the on window. Duration is per condition; no actuator waveform.

REQUIRED OPTICAL SETUP

PD on selected IN1/IN2 with DC coupling and unchanged light/gain in both conditions. Keep PAX physically mounted and optics fixed. Motor-off means rotation stopped, not power unplugged. Outputs hold the configured static biases (default zero).

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

top to bottom: PAX -> 10mm low profile mount stack constrained by bracing arm -> magnetic base unconstrained floating on -> metal plate floating on -> 3 rubber screw caps placed near the edges of the metal plate, all sitting directly on the -> optical breadboard

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

DATA

51 recorded samples; pax_on: 24, pax_off: 27

Source: data.csv; full acquisition settings: recipe.json; status: run.json

PAX motor noise — variance, band [5.0, 1000.0] Hz

PAX_ON

sample_count: 393216
acquired_s: 25.769803775999996
wall_span_s: 30.080699698999524
sampling_rate_hz: 15258.789062499998
dark_corrected_mean_v: 0.003081211509803931
normalized_variance: 0.01338884506337873
normalized_band_variance: 0.00023829329407417958
status: ok

PAX_OFF

sample_count: 442368
acquired_s: 28.991029247999993
wall_span_s: 30.53650931699667
sampling_rate_hz: 15258.789062499998
dark_corrected_mean_v: 0.00417592845581196
normalized_variance: 0.011109998677314891
normalized_band_variance: 0.0009591993018937189
status: ok

COMPARISON

normalized_variance_on_minus_off: 0.002278846386063838
normalized_variance_on_over_off: 1.2051167108342626
normalized_band_variance_on_minus_off: -0.0007209060078195393
normalized_band_variance_on_over_off: 0.24842938647236726
interpretation: Signed on-minus-off excess, not proof of mechanical causation. Compare the same PD bandwidth, optical alignment, light level and gain. Sequential drift, shot noise and electronics can also differ. PAX power/angles have a much slower bandwidth.

PAX ON: STATIC POLARIZATION / POWER

sample_count: 192
mean_power_w: 0.0010836203229166667

PAX motor noise — variance, band [5.0, 1000.0] Hz

normalized_power_variance: 0.0006050537678513537

u_variance_rad2: 0.022969563563067105

v_variance_rad2: 0.01848130740045958

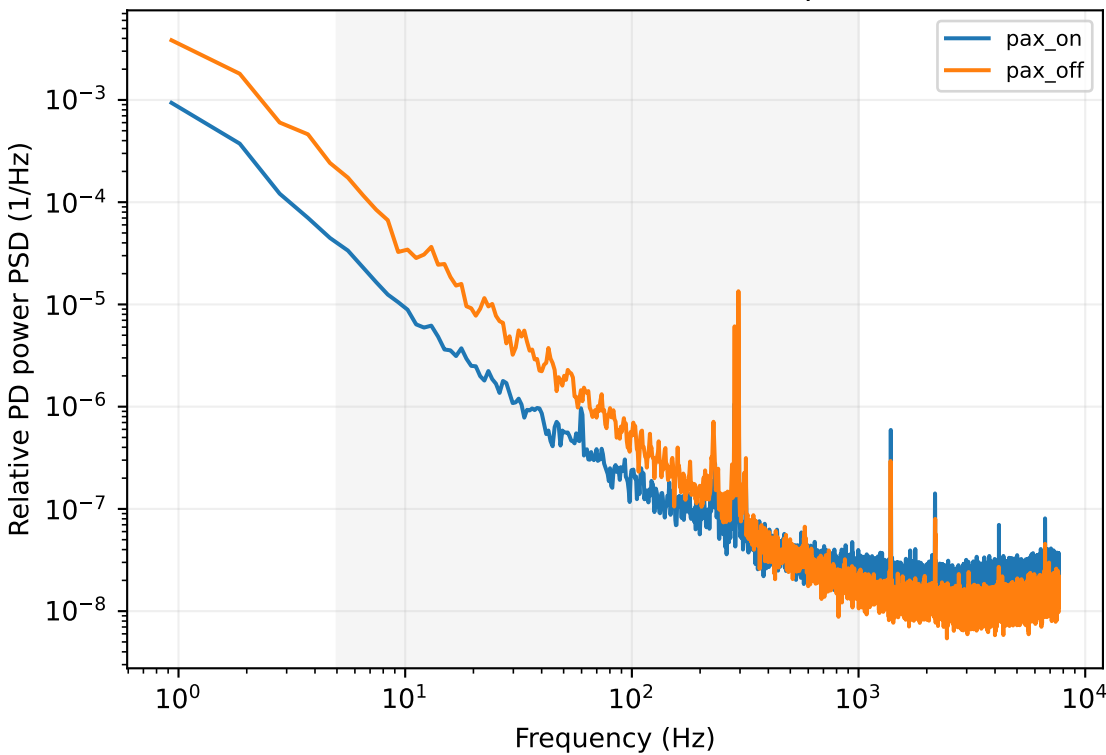
mean_dop: 0.9232120406250001

minimum_sin_v: 0.9558467152795432

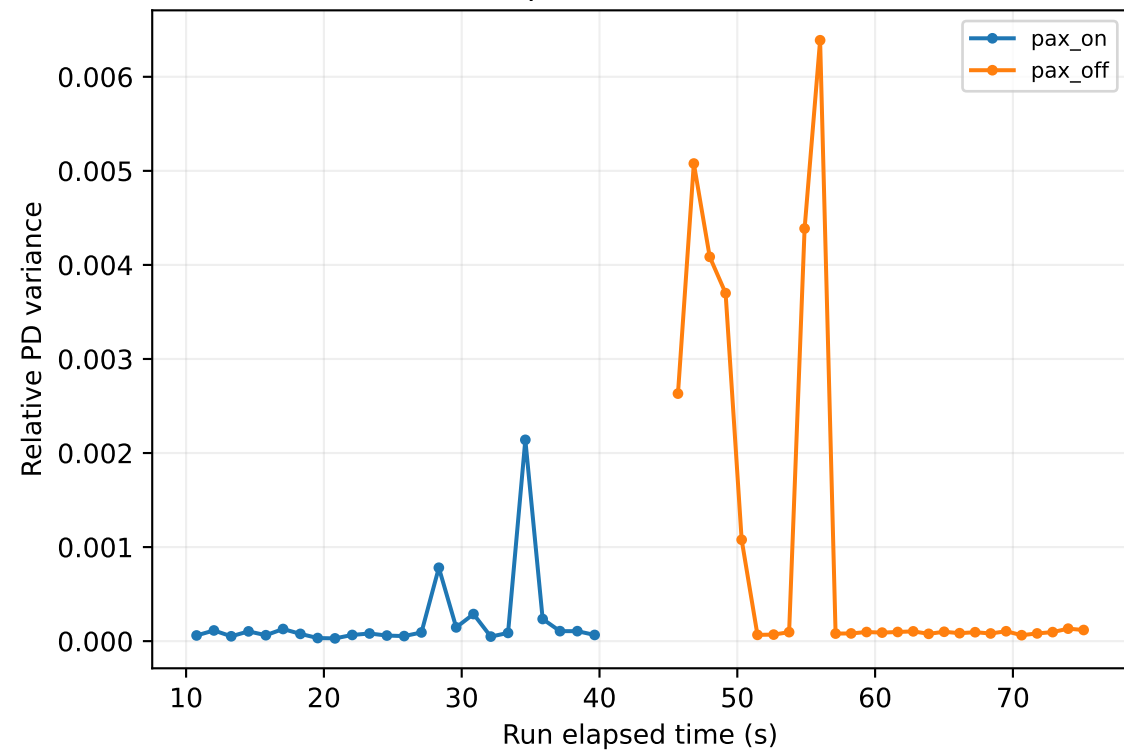
angle_note: u unwrapped in acquisition order; coordinate noise grows near S1 poles

update_rate_hz: 6.384774655891962

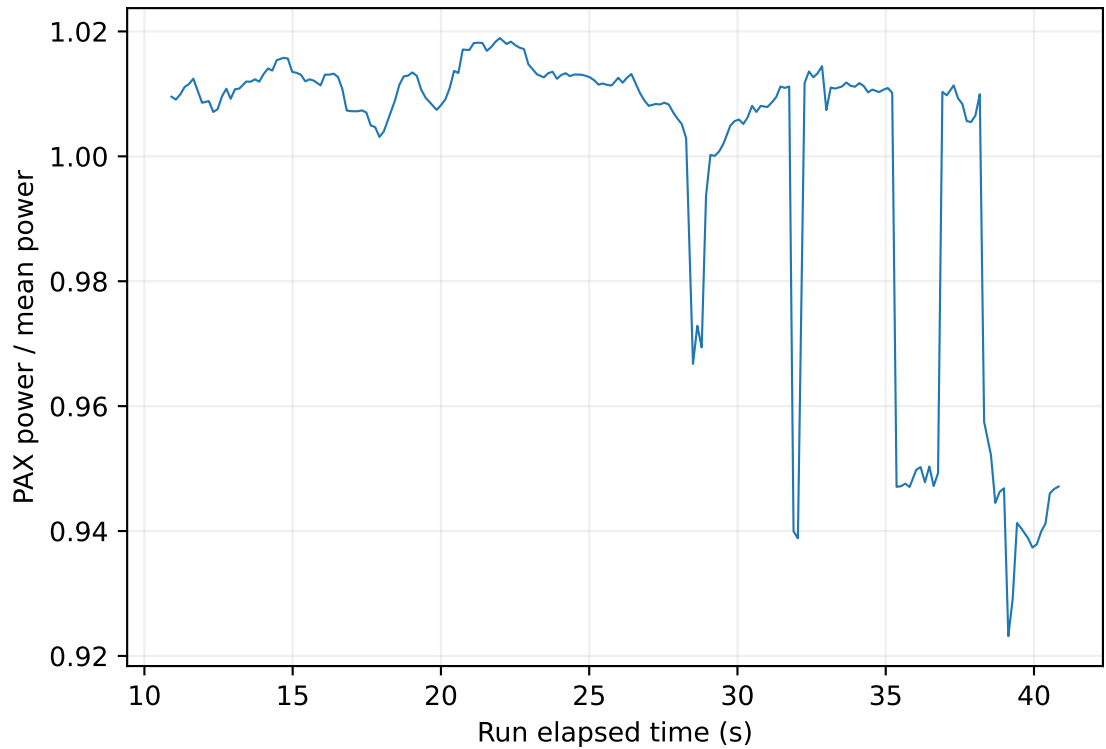
Identical PD bandwidth; shaded comparison band



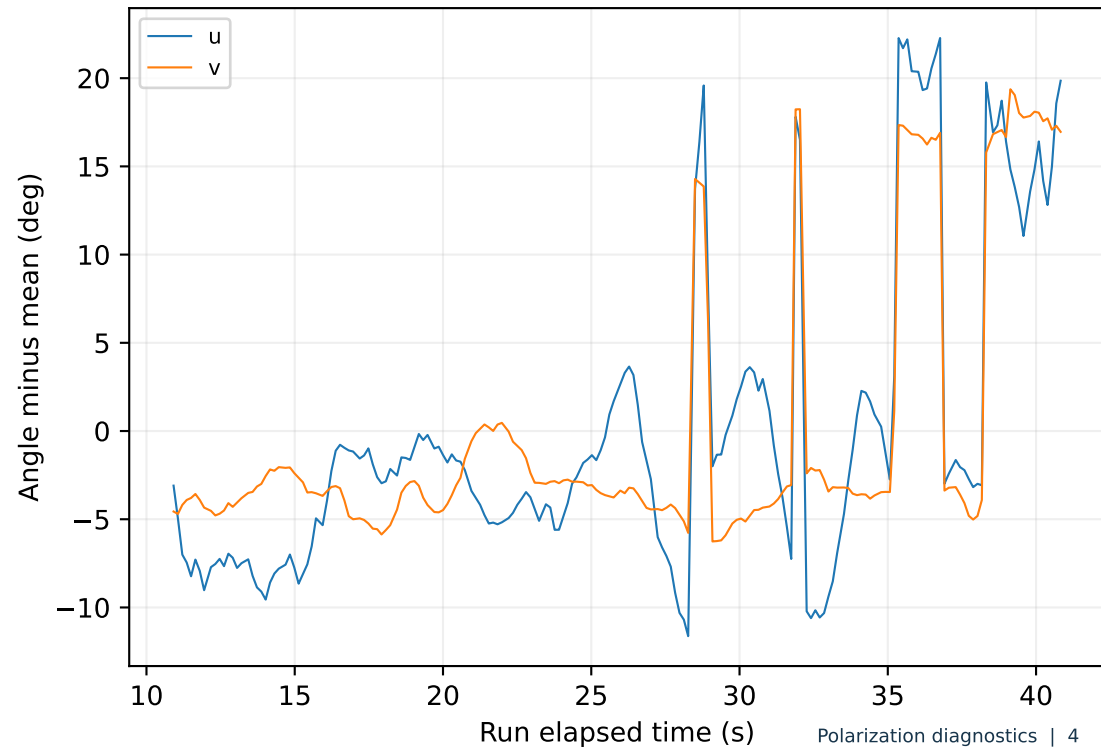
Per-capture band variance



PAX on: slow reference only



Static S1-polar sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"rubber-tip-base"
comment	"top to bottom: PAX -> 10mm low profile mount stack constrain
duration_s	30.0
report	"pdf"
pd_input	"in2"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
pax_vibration_scope_decimation	8192
pax_vibration_band_low_hz	5.0
pax_vibration_band_high_hz	1000.0
pax_vibration_motor_settle_s	5.0
pax_vibration_dark_voltage_v	null
pax_vibration_phi1_bias_voltage	0.0
pax_vibration_phi2_bias_voltage	0.0